

Ramifications in curves (Riemann surfaces)

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October 16, 2025

If a holomorphic map $f: C \rightarrow D$ of compact Riemann surfaces has no ramification points, then the induced map of their fundamental group is injective.

So for the surjectivity of the fundamental group of the induced map, it is necessary to ramify. But being ramified alone does not imply surjectivity. Those maps where the induced map has the surjectivity of the fundamental group are called genuinely ramified. In this talk, we will show many equivalent conditions for genuine ramification. It is shown that the genuine ramification is equivalent to:

- (1) $\pi_1(C) \rightarrow \pi_1(D)$ of the étale fundamental group is surjective.
- (2) The fibre product $C \times_D C$ of C with itself over D is connected.
- (3) The Beta-subbundle of the direct image of the structure sheaf is the structure sheaf.
- (4) The maximal subsheaf on which the natural connection is regular is the structure sheaf.

We will also discuss how to generalize this to positive characteristics and higher dimensions. The major part of the discussion is joint work with Indranil Biswas and Manish Kumar.